INDEPENDENT UNDERGRADUATE RESEARCH PLAN

Five-Finger Tendon-Driven Robotic Hand

A 20-Week Academic Learning, Engineering, and Experimental Research Handbook

Proposed study: Experimental Evaluation of Contact-Feedback Control Strategies for a Low-Cost Five-Finger Tendon-Driven Robotic Hand.

Research platform: Existing mechanical CAD/STL and baseline six-servo actuation architecture from the JulianLee-07/Robotic-Hand repository, contingent upon technical verification and appropriate licensing.

Independent contribution: Five-channel contact sensing, real-time bounded feedback controllers, traceable data acquisition, controlled grasp experiments, statistical analysis, and reproducible reporting.

Planned duration: 20 weeks | Researcher: 1 | Workload: 8–10 h/week (160–200 h total)

Planning baseline: October 2026. This handbook specifies a prospective study; it does not represent completed experiments or validated hardware performance.

1. Research Framework

1.1 Research question

Under a fixed low-cost five-finger tendon-driven hardware platform, does fingertip contact feedback improve 10-second grasp retention relative to fixed closure, and how does it affect observed fingertip contact loading and response to controlled disturbances?

1.2 Research hypotheses

- H1: Contact-stop or bounded contact-maintenance control improves retention under specified test conditions relative to M0 fixed closure.
- H2: Feedback-based strategies reduce unnecessary contact-signal magnitude at comparable retention, subject to calibration limitations.
- H3 (exploratory): Performance differences depend on object geometry and externally imposed disturbances.

1.3 Predefined comparison conditions

- M0—Fixed closure: predetermined actuator commands; FSRs are recorded but not used for normal regulation.
- M1—Contact stop: individual fingers stop increasing closure after persistent threshold exceedance.
- M2—Contact maintenance: bounded, filtered proportional updates follow initial contact to maintain preselected contact targets.

1.4 Measured outcomes

- Primary: predeclared grasp-retention success over a 10-second unsupported hold, incorporating stated displacement and fault rules.
- Secondary: observed displacement, contact-signal peaks and hold-window values, closure-to-contact time, fault counts, and documented failure modes.
- Force in newtons may be reported only after adequate installed-sensor calibration. Otherwise, raw ADC counts or normalized contact units will be used.

1.5 Study limitations

No claims of fully autonomous grasp planning, independent joint-level torque control, general in-hand manipulation, slip detection solely from FSR thresholding, or human-subject performance are within scope. The source servo already implements an internal position loop; this study adds a supervisory contact-feedback loop.

2. Execution and Governance

The project is a single-researcher, sequential integration study. Each week contains directed learning, experimental/development tasks, artifacts, and a stop/go acceptance gate.

2.1 Time allocation

- Default workload: approximately 2 hours targeted study, 4–5 hours building/programming, 2 hours verification, and 1 hour research log and documentation.
- Weeks 14–18 shift time from construction to trials, validation, analysis and interpretation.
- The original two-researcher handbook prescribed approximately 320–360 team person-hours. The solo version therefore reuses source mechanics and reduces final trial volume, rather than preserving the original workload.

2.2 Critical dependencies

- Before procurement: verify models, parts, license, supply ratings and available ADC channels.
- Before full-hand assembly: demonstrate reliable one-finger motion, contact acquisition and controlled release.
- Before final data: five-channel sensing, valid telemetry, controller safety and consistent object setup must pass checks.
- Before reporting: all valid and invalid trials must be traceable to raw files; missing outcomes cannot be silently excluded.

2.3 Planned trial design

An illustrative feasible target is 3 controllers × 4 held-out objects × 2 conditions (baseline and standardized disturbance) × 5 repeated attempts = 120 valid trials. If M2 is unstable, a prospectively amended M0/M1 study contains 80 valid trials under the same object/condition design. Small cell sizes imply substantial uncertainty; reported intervals and limitations must reflect that.

2.4 Budget and safety

CAD 200–250 is a planning constraint, not a verified quotation. Include shipping, tax, six servos, motor power, sensor wiring, ADC expansion, printed parts and spare tendons. Use a rated independent motor supply, accessible physical disconnect, guarded motion limits and a catch tray. Avoid prolonged servo stalls or powering all servos through the MCU USB rail.

WEEK 01 / 20

Research framing and repository audit

Primary learning objectives

Tendon-drive fundamentals; underactuation; Git fundamentals; scientific hypotheses.

Implementation and research tasks

1. Clone and inventory the source repository, including STL/CAD, firmware, calibration utilities, BOM and wiring instructions.
2. Document the six-servo/five-finger actuation map and trace the command-to-motion pathway.
3. Check file accessibility, licenses and redistribution permissions; create an attribution record.
4. Specify the research question, outcomes, M0/M1/M2 definitions, and scope boundaries.

Evidence and deliverables

docs/repository_audit.md; docs/research_charter.md; provenance register.

Acceptance gate

All critical source files are locatable; licensing uncertainty is recorded; the research question is falsifiable.

Selected open educational resources

R01 JulianLee-07, Robotic-Hand repository
https://github.com/JulianLee-07/Robotic-Hand

R02 Yale OpenHand project
https://www.eng.yale.edu/grablab/openhand/

R03 Software Carpentry, Version Control with Git
https://swcarpentry.github.io/git-novice/

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 02 / 20

Architecture, procurement and safety

Primary learning objectives

PWM position servos; PCA9685 and I2C; power distribution; Arduino pin and ADC constraints.

Implementation and research tasks

1. Inspect printable geometry and assembly documentation before buying the full kit.
2. Prepare a part-specific BOM for six positional servos, actuator supply, five FSRs, ADC expansion, fasteners, line and printing.
3. Choose and justify a five-channel sensing architecture (external ADC, multiplexing or a different MCU).
4. Write a wiring and safety plan with common signal ground, overcurrent protection and physical actuator power isolation.

Evidence and deliverables

hardware/bom_v1.csv; architecture diagram; risk register; staged purchase list.

Acceptance gate

Estimated landed cost is reviewed against CAD 200–250 target; parts are electrically and mechanically plausible.

Selected open educational resources

R01 JulianLee-07, Robotic-Hand repository
https://github.com/JulianLee-07/Robotic-Hand

R04 Arduino reference
https://docs.arduino.cc/language-reference/

R05 Arduino Servo Library
https://docs.arduino.cc/libraries/servo/

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

Single-servo characterization

Primary learning objectives

Arduino C++ basics; timer-based nonblocking operation; servo positioning and endpoint calibration.

Implementation and research tasks

1. Bring up one servo without tendon load; characterize usable motion and check for continuous-rotation mismatch.
2. Run or adapt upstream calibration utilities; log safe angle/pulse endpoints and neutral direction.
3. Test supply under motion; record current, stalls and reset events when measurement equipment permits.
4. Create a minimal OPEN/CLOSE/STOP command demonstration.

Evidence and deliverables

firmware/servo_bringup; calibration/servo_01.csv; short test video.

Acceptance gate

Repeated unloaded commands move in the expected direction without hard-stop contact or MCU reset.

Selected open educational resources

R01 JulianLee-07, Robotic-Hand repository
https://github.com/JulianLee-07/Robotic-Hand

R05 Arduino Servo Library
https://docs.arduino.cc/libraries/servo/

R06 Arduino, Blink Without Delay
https://docs.arduino.cc/built-in-examples/digital/BlinkWithoutDelay/

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 04 / 20

Build and verify one finger

Primary learning objectives

Tendon routing; backlash; friction; spring return; basic sensor circuits and voltage dividers.

Implementation and research tasks

1. Print and assemble one finger using the verified source geometry.
2. Route tendon and return element, then characterize slack, range and repeatability.
3. Mount one FSR under a reproducible contact cap, and test unloaded and lightly loaded ADC output.
4. Record mechanical faults and revise assembly instructions without silently modifying source licenses.

Evidence and deliverables

hardware/finger_01_log.md; wiring diagram; photos; raw ADC trace.

Acceptance gate

Twenty low-speed motion cycles complete with no tangles; sensing stays within ADC limits.

Selected open educational resources

R01 JulianLee-07, Robotic-Hand repository
https://github.com/JulianLee-07/Robotic-Hand

R07 Interlink Electronics, FSR integration guide
https://www.interlinkelectronics.com/fsr-402

R08 Arduino Analog Input
https://docs.arduino.cc/built-in-examples/analog/AnalogInput/

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 05 / 20

Python–Arduino experimental interface

Primary learning objectives

pySerial; CSV schemas; timestamps; parsing; functions and exceptions.

Implementation and research tasks

1. Create versioned serial commands and acknowledgments; distinguish commanded angle from measured position.
2. Log raw FSR samples, controller state, servo commands, timestamps and error flags.
3. Create a synthetic-device test path and malformed-packet tests.
4. Record communication dropout behavior and emergency stop semantics.

Evidence and deliverables

docs/interface_v1.md; host/logger.py; synthetic example CSV.

Acceptance gate

A 60-second log can be parsed and plotted; bad inputs and disconnections are handled predictably.

Selected open educational resources

R03 Software Carpentry, Version Control with Git
https://swcarpentry.github.io/git-novice/

R09 pySerial documentation
https://pyserial.readthedocs.io/en/latest/

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 06 / 20

Sensor calibration and signal analysis

Primary learning objectives

pandas, NumPy and Matplotlib; noise, drift, hysteresis; normalization.

Implementation and research tasks

1. Collect repeatable zero, ascending-load and descending-load measurements with a mounted FSR.
2. Measure a held-load segment and investigate remount sensitivity.
3. Plot raw readings, filtered readings and loading/unloading traces.
4. Specify a calibration validity region; use normalized contact units if force conversion is not defensible.

Evidence and deliverables

calibration/fsr_01_raw.csv; analysis/calibration.ipynb or .py; plots.

Acceptance gate

Installed-sensor readings are monotonic in an intended operating region and artifacts are reproducible.

Selected open educational resources

R07 Interlink Electronics, FSR integration guide
https://www.interlinkelectronics.com/fsr-402

R10 pandas getting started
https://pandas.pydata.org/docs/getting_started/

R11 Matplotlib pyplot tutorial
https://matplotlib.org/stable/tutorials/pyplot.html

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 07 / 20

Single-finger closed-loop foundation

Primary learning objectives

Finite-state machines; debouncing; hysteresis; fault conditions and timeouts.

Implementation and research tasks

1. Implement constrained OPEN, APPROACH, CONTACT, HOLD, RELEASE and FAULT states.
2. Implement M0 single-finger fixed closure and M1 contact stop with independent limits.
3. Log contact detection, threshold crossing and watchdog events.
4. Compare repeated open-loop and contact-stop sequences using soft test objects.

Evidence and deliverables

firmware/state_machine_v1; docs/control_logic.md; experimental traces.

Acceptance gate

Twenty contact-and-release cycles are logged; contact stop activates on sensor evidence rather than time alone.

Selected open educational resources

R06 Arduino, Blink Without Delay
https://docs.arduino.cc/built-in-examples/digital/BlinkWithoutDelay/

R07 Interlink Electronics, FSR integration guide
https://www.interlinkelectronics.com/fsr-402

R09 pySerial documentation
https://pyserial.readthedocs.io/en/latest/

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 08 / 20

Scale the mechanical platform to five fingers

Primary learning objectives

Mechanical alignment; multi-axis actuation; power-supply sizing.

Implementation and research tasks

1. Print and assemble the remaining fingers, thumb rotation and bench fixture.
2. Verify six actuator channels individually before synchronized commands.
3. Document per-finger tendon preload, mechanical travel and contact-pad coverage.
4. Check stable operation with all channels active at conservative speeds.

Evidence and deliverables

hardware/assembly_v1; six-servo calibration table; build photos.

Acceptance gate

All five fingers operate independently; no collision, binding or supply-reset fault is ignored.

Selected open educational resources

R01 JulianLee-07, Robotic-Hand repository
https://github.com/JulianLee-07/Robotic-Hand

R05 Arduino Servo Library
https://docs.arduino.cc/libraries/servo/

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 09 / 20

Five-channel sensing and data acquisition

Primary learning objectives

ADC sampling; sample synchronization; sensor baseline management.

Implementation and research tasks

1. Install five contact sensors and justify wiring and ADC choice.
2. Extend serial telemetry, calibration IDs and configuration to all five channels.
3. Confirm each sensor maps to the correct finger and command channel.
4. Record multi-finger contact traces and characterize cross-talk or noisy cables.

Evidence and deliverables

firmware/five_channel_acquisition; host/validation.py; sensor map.

Acceptance gate

All five sensors log without systematic channel mix-ups, significant saturation or unacceptable packet loss.

Selected open educational resources

R07 Interlink Electronics, FSR integration guide
https://www.interlinkelectronics.com/fsr-402

R09 pySerial documentation
https://pyserial.readthedocs.io/en/latest/

R10 pandas getting started
https://pandas.pydata.org/docs/getting_started/

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 10 / 20

M0: reproducible open-loop baseline

Primary learning objectives

Experimental baselines; parameter freezing; automated trial summaries.

Implementation and research tasks

1. Implement a fixed six-channel closure trajectory independent of contact readings.
2. Keep the same safety bounds that will apply to M1 and M2.
3. Construct a repeatable placement jig and define the baseline object set.
4. Run 20 consecutive development-object attempts and preserve failures.

Evidence and deliverables

configs/m0_v1.yaml; data/pilot_m0; baseline reliability report.

Acceptance gate

The baseline procedure runs continuously with a complete command, sensing and outcome record.

Selected open educational resources

R10 pandas getting started
https://pandas.pydata.org/docs/getting_started/

R12 Software Carpentry, Plotting and Programming in Python
https://swcarpentry.github.io/python-novice-gapminder/

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 11 / 20

M1: independent contact-stop control

Primary learning objectives

Threshold estimation; filtering, hysteresis and individual finger state transitions.

Implementation and research tasks

1. Implement per-finger contact stop with persistence across consecutive samples.
2. Add closure/travel/time/contact upper bounds and sensor-fault response.
3. Use only development objects for threshold tuning.
4. Compare M0 and M1 on the same initial conditions and inspect first-contact times.

Evidence and deliverables

configs/m1_v1.yaml; comparison traces; threshold log.

Acceptance gate

Every finger stops on its own contact evidence; bounds stay active for all methods.

Selected open educational resources

R07 Interlink Electronics, FSR integration guide
https://www.interlinkelectronics.com/fsr-402

R12 Software Carpentry, Plotting and Programming in Python
https://swcarpentry.github.io/python-novice-gapminder/

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 12 / 20

M2: bounded contact-maintenance controller

Primary learning objectives

Proportional feedback; deadband; gain, rate limits and sample timing.

Implementation and research tasks

1. Implement an outer-loop P update after initial contact, without claiming direct torque control.
2. Start from conservative gains; log measured error, filtered signal, command and saturation.
3. Measure settling, hunting, drift and 10-second hold behavior under safe disturbances.
4. Use identical grasp targets and safety limits when comparing M1 to M2.

Evidence and deliverables

firmware/m2_controller; configs/m2_candidate.yaml; gain sweep plots.

Acceptance gate

M2 does not systematically creep into an endpoint or persistently oscillate during hold; else reduce scope to M0/M1.

Selected open educational resources

R06 Arduino, Blink Without Delay
https://docs.arduino.cc/built-in-examples/digital/BlinkWithoutDelay/

R13 University of Michigan, PID Control
https://ctms.engin.umich.edu/CTMS/index.php?example=Introduction&section;=ControlPID

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 13 / 20

Integrated tuning and controlled pilot

Primary learning objectives

Design of experiments; operator effects; repeatability and pilot interpretation.

Implementation and research tasks

1. Test all retained controllers on development-only objects with repeated placements.
2. Measure trial duration, mechanical wear, sensor drift and video visibility.
3. Estimate attainable formal trial count within remaining hours.
4. Write unambiguous definitions of valid failure, invalid recording, displacement and excessive deformation.

Evidence and deliverables

data/pilot; docs/pilot_report.md; proposed experimental SOP.

Acceptance gate

A recorded trial can be repeated from the SOP and classified consistently by rewatching the video.

Selected open educational resources

R10 pandas getting started
https://pandas.pydata.org/docs/getting_started/

R12 Software Carpentry, Plotting and Programming in Python
https://swcarpentry.github.io/python-novice-gapminder/

R14 Center for Open Science, Preregistration
https://www.cos.io/initiatives/prereg

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 14 / 20

Freeze the experimental protocol

Primary learning objectives

Randomized balanced blocks; preregistration; outcome operationalization.

Implementation and research tasks

1. Select four final objects separately from tuning objects and document geometry/surface/mass.
2. Predeclare 3 methods x 4 objects x 2 conditions x 5 repeats = 120 valid attempts if M2 passes; revise down prospectively if needed.
3. Randomize method order within object-condition blocks and freeze configuration, seed and definitions.
4. Choose an explicit disturbance event, 10-second unsupported hold, success criteria and replacement rules.

Evidence and deliverables

docs/protocol_v1.md; schedule.csv; frozen Git tag; object catalog.

Acceptance gate

No final-object parameter tuning is allowed after protocol freeze; every metric has a defined measurement rule.

Selected open educational resources

R14 Center for Open Science, Preregistration
https://www.cos.io/initiatives/prereg

R15 Scientific practice: predeclared hypothesis and metrics
https://www.cos.io/initiatives/prereg

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 15 / 20

Formal data collection: first balanced batch

Primary learning objectives

Research data stewardship; validation of run metadata; reproducibility.

Implementation and research tasks

1. Execute approximately half of frozen scheduled attempts with standardized alignment and support removal.
2. Start/end sessions with zero/reference sensor checks and actuator inspection.
3. Capture synchronized video IDs and log all failures, aborts and invalid recordings.
4. Create immediate checks for missing data, duplicate IDs and configuration mismatches.

Evidence and deliverables

data/raw/session_01...; videos manifest; quality dashboard.

Acceptance gate

Every scheduled attempt maps to a log, video, method, configuration, object and outcome.

Selected open educational resources

R10 pandas getting started
https://pandas.pydata.org/docs/getting_started/

R12 Software Carpentry, Plotting and Programming in Python
https://swcarpentry.github.io/python-novice-gapminder/

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 16 / 20

Complete formal data collection

Primary learning objectives

Reliability audits; measurement drift; systematic failure taxonomy.

Implementation and research tasks

1. Complete the remaining balanced attempts; do not repeat valid failures for a better result.
2. Record wear, service events and protocol deviations without obscuring altered hardware.
3. Audit success labels with recorded videos, preserving the predeclared rubric.
4. Lock raw data; prepare analysis-ready table and invalid-attempt ledger.

Evidence and deliverables

data/raw locked manifest; data/processed/trials.csv; audit report.

Acceptance gate

All planned valid attempts or transparent shortfalls are reconciled; raw files are unchanged.

Selected open educational resources

R10 pandas getting started
https://pandas.pydata.org/docs/getting_started/

R12 Software Carpentry, Plotting and Programming in Python
https://swcarpentry.github.io/python-novice-gapminder/

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

Quantitative performance analysis

Primary learning objectives

Wilson intervals; categorical outcomes; grouped summaries; visualization.

Implementation and research tasks

1. Compute per-method/object/condition retention counts, success rates and 95% Wilson intervals.
2. Plot contact traces, peak and hold-window contact summaries with failure counts.
3. Analyze time-to-contact, early drops, sensor faults and slip categories where measurable.
4. Avoid treating within-trial sensor samples as independent trials.

Evidence and deliverables

analysis/reproduce_results.py; figures/fig_01...; tables with denominators.

Acceptance gate

All reported results are regenerated from saved data without manual edits.

Selected open educational resources

R10 pandas getting started
https://pandas.pydata.org/docs/getting_started/

R11 Matplotlib pyplot tutorial
https://matplotlib.org/stable/tutorials/pyplot.html

R16 SciPy binomial confidence intervals
https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.binomtest.html

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 18 / 20

Robustness and uncertainty assessment

Primary learning objectives

Control robustness; sensitivity analysis; failure interpretation and limitations.

Implementation and research tasks

1. Compare performance under the predeclared no-disturbance and disturbance conditions.
2. Examine effects of sensor drift, mechanical slack and between-session variation.
3. Distinguish contact-signal reduction from actual reduction in physical gripping force.
4. Explain trade-offs among retention, contact signal and completion time; document inconclusive findings.

Evidence and deliverables

analysis/robustness.py; failure mechanism matrix; results memo.

Acceptance gate

Conclusions match the evidence and acknowledge low sample size, object diversity and calibration limits.

Selected open educational resources

R10 pandas getting started
https://pandas.pydata.org/docs/getting_started/

R13 University of Michigan, PID Control
https://ctms.engin.umich.edu/CTMS/index.php?example=Introduction&section;=ControlPID

R16 SciPy binomial confidence intervals
https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.binomtest.html

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 19 / 20

Academic manuscript and independent reproducibility

Primary learning objectives

Scientific writing; IEEE-style citations; versioned environments.

Implementation and research tasks

1. Draft a 6–8 page technical report with abstract, background, methods, results, discussion and limitations.
2. Write original-hardware attribution and explain adaptations and independent contributions.
3. Recreate figures on a clean Python environment and verify a fresh startup checklist.
4. Document negative results and avoid unsupported novelty or force-precision claims.

Evidence and deliverables

report/manuscript_v1.pdf; requirements lockfile; replication checklist.

Acceptance gate

A clean machine regenerates figures and a reader can trace every quantitative claim to data.

Selected open educational resources

R01 JulianLee-07, Robotic-Hand repository
https://github.com/JulianLee-07/Robotic-Hand

R03 Software Carpentry, Version Control with Git
https://swcarpentry.github.io/git-novice/

R17 GitHub Releases documentation
https://docs.github.com/en/repositories/releasing-projects-on-github/managing-releases-in-a-repository

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

WEEK 20 / 20

Verification, release and research presentation

Primary learning objectives

Research communication; artifact archival; technical limitations and future work.

Implementation and research tasks

1. Run a final independent reproduction using the documented controls on development objects.
2. Produce a 2–3 minute demo with labeled methods and representative failure cases.
3. Publish only material whose permission is clear; archive code, analysis, metadata and report.
4. Prepare a 10-minute research talk and a concise contribution statement.

Evidence and deliverables

release v1.0; demonstration video; final report; complete reproducibility archive.

Acceptance gate

A third party can follow the build/experiment documentation and regenerate the primary figures.

Selected open educational resources

R01 JulianLee-07, Robotic-Hand repository
https://github.com/JulianLee-07/Robotic-Hand

R17 GitHub Releases documentation
https://docs.github.com/en/repositories/releasing-projects-on-github/managing-releases-in-a-repository

Weekly research log: hours budgeted/used; assumptions; changes; failed tests; raw evidence locations; next decision.

Appendix A. Experimental and Data Integrity Protocol

Single-trial standard operating procedure

- Confirm the method, object ID, trial ID, operator, hardware configuration, sensor calibration and software commit.
- Start synchronized serial logging and video; verify unique identifiers.
- Position the object with the documented jig; open the hand; execute the assigned controller with identical safety limits.
- Remove support as specified and apply a predefined disturbance only in disturbance trials.
- Observe the full ten-second hold and record retention, displacement, deformation, aborts, and failure category.
- Save the outcome regardless of success. Preserve invalid recordings with explicit reasons and predeclared replacement links.

Data schema

trial_id, replacement_for, session, object_id, condition, controller, random_order, hardware_revision, firmware_commit, config_id, calibration_id, sample_time, raw_adc_1...5, filtered_contact_1...5, command_1...6, state, fault_code, video_file, retention_success, displacement, failure_type, invalid_reason.

Analytic safeguards

- Freeze model and tuning parameters before final-object testing.
- Aggregate by trial, not individual high-frequency samples, when reporting success rates.
- Use 95% Wilson intervals for binomial success estimates and show denominators.
- Do not infer reduced physical gripping force from unvalidated FSR contact signals.
- Keep pilot/development and formal held-out evaluation datasets separate.

Appendix B. Reproducible Project Structure

Suggested version-controlled directory layout

docs/ — research charter, literature notes, operating procedure, engineering log
hardware/ — assembly notes, component manifest, wiring, licensed CAD references
firmware/ — Arduino controller and library version record
host/ — Python serial transport, logger and validation utilities
configs/ — actuator limits, controller parameter sets and schedule seed
calibration/ — source measurements and sensor validity documentation
data/raw/ — immutable telemetry and trial files
data/processed/ — analysis-ready tabular data
analysis/ — scripts that regenerate tables and figures
figures/ — derived visuals with script provenance
report/ — manuscript, citations and supporting appendices

Research reporting structure

- Abstract; Introduction and related work; Hardware and sensor integration; Controller methods; Experimental protocol; Results; Discussion and limitations; Conclusion; References.
- A 6–8 page core technical report with appendices is an appropriate target for this undergraduate project; report quality is more important than page count.
- All source materials must be acknowledged, and externally authored CAD or code may only be redistributed under applicable permissions.

References and Open Educational Resources

The materials below are learning and engineering references. Repository content, hardware specifications and licensing should be rechecked at implementation time; citing a repository does not establish permission to republish its files.

[R01] JulianLee-07, Robotic-Hand repository. https://github.com/JulianLee-07/Robotic-Hand

[R02] Yale OpenHand project. https://www.eng.yale.edu/grablab/openhand/

[R03] Software Carpentry, Version Control with Git. https://swcarpentry.github.io/git-novice/

[R04] Arduino reference. https://docs.arduino.cc/language-reference/

[R05] Arduino Servo Library. https://docs.arduino.cc/libraries/servo/

[R06] Arduino, Blink Without Delay. https://docs.arduino.cc/built-in-examples/digital/BlinkWithoutDelay/

[R07] Interlink Electronics, FSR integration guide. https://www.interlinkelectronics.com/fsr-402

[R08] Arduino Analog Input. https://docs.arduino.cc/built-in-examples/analog/AnalogInput/

[R09] pySerial documentation. https://pyserial.readthedocs.io/en/latest/

[R10] pandas getting started. https://pandas.pydata.org/docs/getting_started/

[R11] Matplotlib pyplot tutorial. https://matplotlib.org/stable/tutorials/pyplot.html

[R12] Software Carpentry, Plotting and Programming in Python. https://swcarpentry.github.io/python-novice-gapminder/

[R13] University of Michigan, PID Control.
https://ctms.engin.umich.edu/CTMS/index.php?example=Introduction&section;=ControlPID

[R14] Center for Open Science, Preregistration. https://www.cos.io/initiatives/prereg

[R15] Scientific practice: predeclared hypothesis and metrics. https://www.cos.io/initiatives/prereg

[R16] SciPy binomial confidence intervals. https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.binomtest.html

[R17] GitHub Releases documentation.
https://docs.github.com/en/repositories/releasing-projects-on-github/managing-releases-in-a-repository

Basis of adaptation

This solo handbook adapts the supplied 34-page two-researcher execution handbook, retaining the Researcher B-centered learning trajectory (Python, Arduino, serial telemetry, sensor processing, control and experimental analysis) and adding necessary independent mechanical/sensing integration. The five-finger platform, six-actuator arrangement, and proposed 120-trial feasibility design are prospective adaptations and not findings reported by that source.